MILITARY SPECIFICATION

TRANSMITTERS, PRESSURE, SYNCHRO, AIRCRAFT, 320-DEGREE MOVEMENT, HIGH TEMPERATURE (177°C), GENERAL PURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers hermetically sealed 320-degree movement synchro type aircraft pressure transmitters for high-temperature installation.

1.2 Classification - The transmitters shall be furnished in the following ranges as specified (see 6.2):

MS28131-1T	MS28131-20D	0- 10 PSI	Differential pressure
MS28131-2T	MS28131-20	0- 10 PSI	Gage pressure
MS28131-3T	MS28131-21D	0- 25 PSI	Differential pressure
MS28131-4T	MS28131-21	0- 25 PSI	Gage pressure
MS28131-5T	MS28131-22D	0- 50 PSI	Differential pressure
MS28131-6T	MS28131-22	0- 50 PSI	Gage pressure
MS28131-7T	MS28131-23D	0- 100 PSI	Differential pressure
MS28131-8T	MS28131-23	0- 100 PSI	Gage pressure
_____	MS28131-24D	0- 150 PSI	Differential pressure
_____	MS28131-24	0- 150 PSI	Gage pressure
_____	MS28131-25D	0- 200 PSI	Differential pressure
_____	MS28131-25	0- 200 PSI	Gage pressure
MS28131-9T	MS28131-26D	0- 250 PSI	Differential pressure
MS28131-10T	MS28131-26	0- 250 PSI	Gage pressure
MS28131-11T	MS28131-27D	0- 300 PSI	Differential pressure
MS28131-12T	MS28131-27	0- 300 PSI	Gage pressure
MS28131-13T	_____	0- 500 PSI	Differential pressure
MS28131-14T	MS28131-28	0- 500 PSI	Gage pressure
MS28131-15T	MS28131-29	0- 1,000 PSI	Gage pressure
MS28131-16T	MS28131-30	0- 1,000 PSI	Approximate gage pressure
MS28131-17T	MS28131-31	0- 2,500 PSI	Approximate gage pressure
MS28131-18T	MS28131-32	0- 5,000 PSI	Approximate gage pressure

FSC 6620

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

BB-N-411	Nitrogen
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-L-5020	Liquid, Compass, Aircraft
MIL-T-5350	Transmitters, Synchro, Aircraft, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-L-6081	Lubricating Oil, Jet Engine
MIL-I-6856	Indicators, Pressure, Synchro, 1-1/2 Inch
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-S-20708	Synchros, General Specification for
MIL-I-23479	Indicator, Pressure, 1-1/2 Inch, Integrally Lighted
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of US Military Property

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

STANDARDS

Military (Continued)

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS28131	Transmitters, Pressure, Synchro, Aircraft, 320-Degree Movement, High Temperature, General Purpose
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - Transmitters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.4).

3.2 Selection of government documents - Except as provided in 3.2.1 and 3.2.2, specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Commercial parts - Commercial parts having suitable properties shall be used, when on the date of invitation for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, and cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.2.3 Part testing - No part shall be used in the equipment without evidence that the part has been tested. The test results may be available from a data exchange program, from previous contracts, or if not so available, the parts must be tested under the specification requirements. Unless required by the contract, the submission of test information on parts is not required, but the contractor shall be able to produce test results when requested by the procuring activity.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.3.3 Metals - Metals shall be corrosion resistant or suitably protected, as specified herein, to resist corrosion due to fuels, salt spray, or atmospheric conditions to which the transmitter may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other, unless protection against electrolytic corrosion is provided.

3.3.4 Protective treatment - When materials are used in the construction of the transmitter that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

3.3.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If the materials are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.6 Fumes and vapors - Materials used in the construction of the transmitter shall not produce corrosive, deleterious, or toxic fumes or vapors under all the conditions specified herein.

3.4 Design and construction - The transmitter shall be in accordance with MS28131 and shall be designed for use with pressure indicators conforming to MIL-I-7057, MIL-I-6856, MIL-I-23479, or as specified by the procuring activity. The electrical characteristics only of the synchro shall conform to MIL-S-20708/6. The rotor of the transmitter synchro shall rotate, relative to the stator, through an angle of 320 $\pm$1 degrees for the range of pressure specified on the applicable MS (see 1.2). The transmitter shall be provided with positive means to adjust or correct the indications. The means of adjustment shall be mechanical and of such nature that adjustment can be accomplished from the exterior of, and without damage to, the hermetically sealed case. The adjustment shall provide a minimum of $\pm$7 degrees of indicator adjustment. The adjustment shall have a positive type locking device and shall be protected from external disturbance except when adjustment is being made. The protection shall be as described in MS28131. The positive type locking device shall be of the No-Drag type which does not affect the transmitter adjustment when locking the adjustment. Transmitters shall be correctly adjusted upon delivery. The transmitting mechanism shall be distinctly separate from the case so that small distortions of the case will not affect the accuracy of the indications. Each transmitter shall be capable of operating either one or two indicators without change, in indication of more than one degree of arc. The transmitter shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service and the environmental conditions of Section 4. The transmitter shall also incorporate a screw type adjustable restrictor for each of the following part numbers:

MS28131-7T	MS28131-23D	MS28131-25D
MS28131-8T	MS28131-23	MS28131-25
MS28131-11T	MS28131-24D	MS28131-27D
MS28131-12T	MS28131-24	MS28131-27

3.4.1 Maintenance - The design shall be such as to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment, which are normally available as commercial standards and readily obtainable at a repair facility.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

3.4.2 Case - The outline dimensions and markings of the transmitter shall conform to MS28131. The case shall be finished in lusterless black material, Color No. 37038 of FED-STD-595.

3.4.2.1 Hermetic sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed, replaced, and the case resealed at least three times. This shall be possible without the use of any special tool, jig, or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any environment to which the transmitters may be subjected in normal use in military aircraft or in environmental tests.

3.4.2.2 Filling medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with type I, class 1, grade C of BB-N-411. The filling medium shall contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 $\pm$0.1 atmosphere.

3.4.2.3 Case pressure release - A metal blow-out device to discharge excessive case pressure shall be provided in each pressure transmitter in the ranges of 0-150 psi through 0-5,000 psi. This device shall operate to release the internal case pressure through an approximately 0.045-inch-diameter orifice whenever the internal case pressure, relative to the atmosphere, reaches 90 $\pm$30 psi. The device shall not affect hermetic sealing. The design of the blow-out device shall be submitted to the procuring activity for prior approval. The case pressure release (CPR) shall be as shown on MS28131.

3.4.3 Differential pressure transmitters - Each differential pressure transmitter shall have two pressure connections, and they shall be marked as shown on MS28131. The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the two pressure connections; and the synchro signal output represent ($P_1 - P_2$). The angular position of the rotor of the synchro relative to its stator shall be essentially a linear function of ($P_1 - P_2$).

3.4.3.1 When a case pressure release is required, the blow-out device shall discharge through a 5/16-24 connection, conforming to MS33656, in an optional location on the back plate of the case. The case pressure release connection shall be marked "CPR". The transmitter shall be so designed and constructed that if either the "P_1" or the "P_2" pressure sensor leaks, the pressure in the case will be relieved, and the escaping fluid discharged through the case pressure release connection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

3.4.4 Gage pressure transmitters - Each gage pressure transmitter shall have two pressure connections and they shall be marked as shown on MS28131. The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the pressure ("P") port and the pressure applied to the vent ("V") port, and the synchro signal output shall represent (pressure $_{p}$ - pressure $_{v}$). The angular position of the rotor of the synchro relative to the stator shall be essentially a linear function of (pressure $_{p}$ - pressure $_{v}$).

3.4.4.1 The 500 and 1,000 psi transmitters shall have a case pressure release constructed so that if the sensor connected to the "P" port should leak, the case pressure would be relieved and the fluid discharged through the vent port. This port shall be marked "V & CPR".

3.4.5 Approximate gage pressure transmitters - Each approximate gage pressure transmitter shall have one pressure connection. The pressure connection shall be marked "P". The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the pressure connection and the pressure in the case of the transmitter, and the synchro signal output shall represent (pressure $_{p}$ - pressure $_{case}$). The angular position of the rotor of the synchro relative to the stator shall be essentially a linear function of (pressure $_{p}$ - pressure $_{case}$).

3.4.5.1 The blow-out device for releasing case pressure shall discharge through a 5/16-24 connection conforming to MS33656. The transmitter shall be so designed and constructed that if the pressure sensor leaks, the pressure in the case will be relieved, and the escaping fluid discharged through the case pressure release connection.

3.4.6 Stops -

3.4.6.1 Overpressure stop (all transmitters) - Each transmitter shall include an overpressure stop which shall act directly on the pressure sensor(s) and which will limit the overtravel of the rotor of the synchro beyond its full-scale position to some value between 1 degree and 15 degrees, inclusive.

3.4.6.2 Reverse pressure stop - Each gage pressure transmitter, up to and including 50 psi, and all differential pressure transmitters shall include a device which will limit the reverse rotation of the rotor of the synchro beyond its zero-pressure position to some value between 1 degree and 30 degrees, inclusive.

3.4.7 Weight - The weight of the completely assembled transmitter shall not exceed 1.4 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

3.4.8 Screw threads - Screw threads, 0.060 inch in diameter or larger, shall be in accordance with MIL-S-7742.

3.4.9 Jewel bearings - When used, synthetic sapphire or ruby jewel bearings shall be in accordance with MIL-B-27497.

3.4.10 Other requirements - The requirements of MIL-T-5350 are applicable as requirements of this specification. Additional requirements are as specified herein. Where the requirements of MIL-T-5350 and this specification conflict, the requirements of this specification shall apply and shall take precedence.

3.5 Electrical requirements -

3.5.1 Primary input power requirements - The transmitter shall operate on ac power (single phase), 26V ±2.0V, 400 Hz ±5 percent, 6.24 va maximum.

3.5.2 Undervoltage protection - The transmitter shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage is within the specified limits.

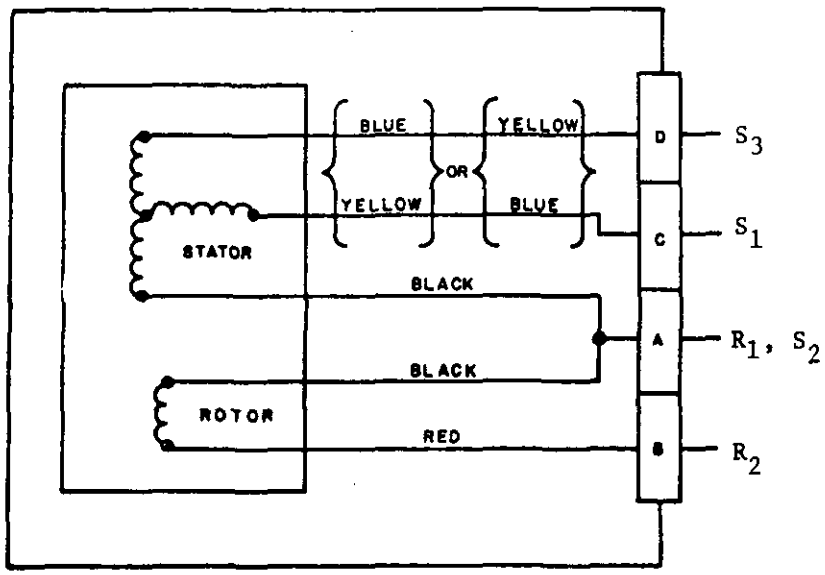
3.5.3 Electronic parts - Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists in accordance with MIL-E-5400 shall not be used, unless approved by the procuring activity.

3.5.4 Electric connectors - The electric connectors shall be as specified on MS28131. These connectors shall be supplied with removable protective plastic cap covers.

3.5.5 Wiring - The internal and external wiring shall be as shown on figures 1 and 2, respectively. Internal wiring shall be color coded, neat, and accomplished in such a manner that individual wires may be easily traced. The wiring shall be insulated from the transmitter case. The signal rotation at the pins of the electrical connector of the transmitter shall be ADC for increasing value of the pressure function being transmitted.

3.5.6 Gears, pivots, joints, and bearings - Before finalizing the design of the pressure transmitter mechanism, the contractor shall investigate and determine the best known types and the proper processing of interfacing materials for use in the gears, pivots, joints, and bearings in the train of moving parts of the transmitter. These interface materials shall be selected for their ability to resist wear, corrosion, and friction when interfacing with one another. The interfacing materials

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.



When the transmitter is connected to a synchro receiver with electrical characteristics in accordance with MIL-S-20708/5, increasing pressure to the transmitter will cause a counterclockwise rotation of the synchro receiver rotor when viewed from the shaft extension end of the synchro.

FIGURE 1. Transmitter internal wiring diagram

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

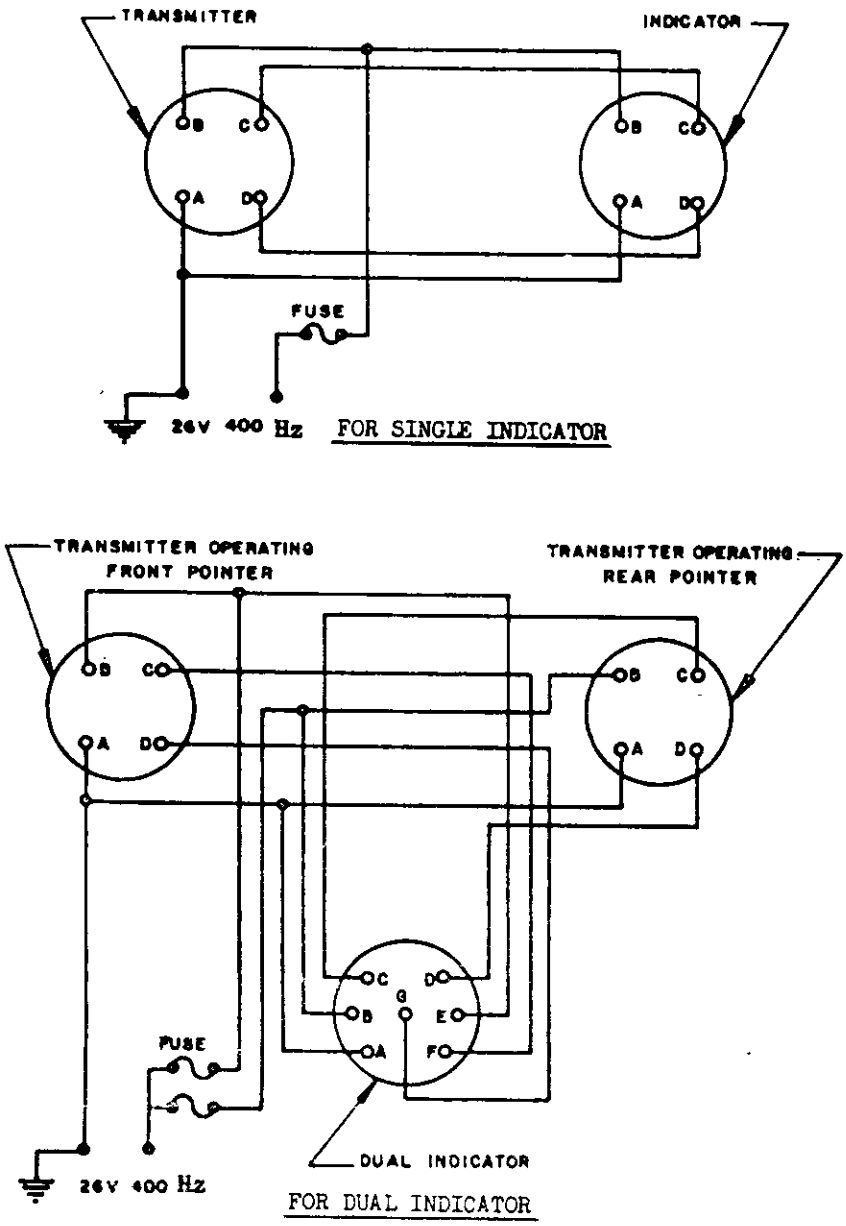


FIGURE 2. External wiring diagrams

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

shall not be dissimilar as described in MIL-STD-889. After selection of the material types and processing procedures, the contractor shall submit a description of each interfacing material and process proposed for the train of moving parts (gears, pivots, joints, and bearings) and present his reason for selection of each type to the procuring activity for approval.

3.6 Performance - The transmitter shall perform satisfactorily when subjected to the tests specified in Section 4.

3.6.1 Reliability in Mean-Time-Between-Failures (MTBF) - The transmitter shall have a minimum acceptable Mean (Operating) Time-Between-Failures of 2000 hours when tested and accepted as outlined under the requirements of 4.4.3 and subparagraphs.

3.7 Identification of product -

3.7.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130.

3.7.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number, including applicable dash numbers, if the drawing is tabulated and covers more than one part.

3.8 Part numbering of interchangeable parts - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.9 Workmanship - Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

4.2 Classification of tests - The inspection and testing of transmitters shall be classified as follows:

a. Qualification inspection (4.3)

b. Quality conformance inspection (4.4)

4.2.1 Precedence - Inspection procedures and inspections shall be as specified herein and in MIL-T-5350. Whenever a procedure required by this specification conflicts with a corresponding procedure required by MIL-T-5350, the procedure required by this specification shall apply.

4.3 Qualification inspection - The qualification inspection of the transmitter shall consist of all the examinations and tests of this specification performed in the order specified under the paragraph headed inspection methods, and samples submitted for Qualification shall be accompanied by the supplier's report showing that the reliability test has been satisfactorily met on other samples of the same equipment.

4.3.1 Qualification inspection sample - Qualification inspection samples shall consist of three transmitters manufactured in accordance with this specification. The transmitters submitted for qualification inspection shall have been previously subjected only to individual inspections. The samples shall be forwarded at the contractor's expense to the laboratory designated in the letter of authorization (see 6.4).

4.3.1.1 Qualification inspection sample identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for qualification inspection
TRANSMITTER, PRESSURE, SYNCHRO, AIRCRAFT, 320-DEGREE MOVEMENT, HIGH TEMPERATURE (177°C), GENERAL PURPOSE
Submitted by (manufacturer's name, date) for qualification inspection in accordance with MIL-T-7748E under authorization (reference letter authorizing tests)
MS part number _____

4.4 Quality conformance inspection - The quality conformance inspection shall consist of the individual inspection, the sampling plans, and the reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required inspections. Sampling plan B shall be conducted at a Government laboratory designated by the procuring activity (see 6.6). Quality conformance inspection, except for sampling plan B, shall be under the supervision of the Government quality control representative. The contractor

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the final acceptance of the finished product.

4.4.1 Individual inspection - Each transmitter submitted for acceptance shall be subjected to the individual inspection. This inspection shall determine compliance with requirements for material, workmanship, operational adequacy, and reliability. As a minimum, each transmitter accepted shall have passed the following inspections:

(a)	Examination of product	(4.6.1)
(b)	Scale error at room temperature	(4.6.2)
(c)	Friction error	(4.6.3)
(d)	Position error	(4.6.4)
(e)	Dielectric strength	(4.6.5)
(f)	Sensors matching (for differential transmitters only)	(4.6.6)
(g)	Proof pressure	(4.6.7)
(h)	Sealing	(4.6.8)

4.4.2 Sampling plans - The sampling plans shall consist of sampling plan A and sampling plan B inspections. The inspection samples selected for sampling tests shall first have passed the individual inspection. The samples which have been subjected to sampling plan A inspection shall not be delivered on contract until they have been refurbished (see 6.3) and resubmitted and passed all the individual inspections. Inspection samples which have been subjected to the sampling plan B inspection shall not be delivered on contract.

4.4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity offered for acceptance	Quantity to be selected for inspection
First 15	(See note)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

NOTE: When sampling plan B is invoked, the quantity shall be zero. When sampling plan B is to be omitted, the quantity shall be one.

When a defective transmitter occurs, no transmitters from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected with the corrections approved by the procuring activity.

4.4.2.1.1 Sampling plan A inspection - Each sample selected shall be subjected to the following inspections in the order listed:

(a)	Damping	(4.6.9)
(b)	Suction	(4.6.10)
(c)	Voltage and frequency variation	(4.6.11)
(d)	Seasoning	(4.6.12)
(e)	High temperature	(4.6.13)
(f)	Low temperature	(4.6.14)
(g)	Vibration resonance search and dwell	(4.6.15.1)

4.4.2.2 Sampling plan B instructions - Two transmitters shall be selected at random, from the first 15 produced on contract, and submitted within 10 days after manufacture. These samples shall be forwarded at the contractor's expense, to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached and marked with the following information:

TRANSMITTER, PRESSURE, SYNCHRO, AIRCRAFT, 320-DEGREE MOVEMENT, HIGH TEMPERATURE (177°C), GENERAL PURPOSE
Submitted by (Manufacturer's name, date)
for quality conformance sampling plan B inspection, in accordance with contract/order No._____
Manufacturer's part number_____
MS Part No. _____

4.4.2.2.1 Sampling plan B inspection - Each sample selected shall be subjected to the following inspections in the order listed:

(a)	Sampling plan A inspection	(4.4.2.1.1)
(b)	Vibration cycling	(4.6.15.2)
(c)	Low temperature exposure	(4.6.16)
(d)	High temperature exposure	(4.6.17)
(e)	Effect of lead length	(4.6.18)
(f)	Radio noise suppression	(4.6.19)
(g)	Electrical characteristics	(4.6.20)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

(h)	Overpressure	(4.6.21)
(i)	Salt fog	(4.6.22)
(j)	Case pressure release and test for helium	(4.6.23)
(k)	Case proof pressure	(4.6.24)
(l)	Internal examination	(4.6.25)
(m)	Torque	(4.6.26)

4.4.3 Reliability assurance tests - Reliability assurance tests shall be conducted in accordance with MIL-STD-781, Qualification Phase and Production Sampling Phase (For Army or Navy Contracts Only) Tests shall be conducted. Equipments selected for the reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Qualification Phase - Prior to Qualified Products Listing (QPL) (see 3.1) a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled, "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be seven. For the Qualification Phase, Test Level J shall be used except the upper temperature level shall be 177°C and the vibration level shall be 10g $\pm$ 10% at 180 to 240 Hz. The accept/reject criteria for Test Plan I shall be used.

4.4.3.2 Reliability Production Sampling Phase (for Army or Navy contracts only) - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled, "Production Acceptance (Sampling) Phase of Production Reliability Tests". For the Sampling Phase Tests, Test Level J shall be used except the upper temperature level shall be 177°C and the vibration level shall be 10g $\pm$ 10% at 180 to 240 Hz. The accept/reject criteria for Test Plan II shall be used.

4.4.3.2.1 Procedure for production acceptance (sampling) phase - The Reliability Production Acceptance (Sampling) Phase Tests shall be conducted at least once for each lot. (The equipments constructed during one month shall constitute a lot). The number of samples shall be in accordance with Table 5 of MIL-STD-781 or as otherwise approved in the test procedures. The test program shall start with the first month's production. The samples shall be tested until an accept or reject decision is reached. Unless otherwise specified in the contract or order, no equipments shall be shipped until an accept decision is reached for each lot. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment quality control, etc.

4.4.3.3 Test details - The test details, such as the length of the test cycle, the length of the heat portion of the temperature cycle, the method of conducting the performance tests, special failure criteria,

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

preventive maintenance to be allowed, etc., shall be part of the test procedures to be submitted to and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests. The following paragraphs shall be considered as minimum requirements and apply to both phases.

4.4.3.3.1 Duty cycle - The duty cycle shall be continuous and shall consist of applying pressure to the transmitter at a rate of 2 to 3 cycles (minimum and maximum) per minute from 0 to 80 percent of full scale and return. The test shall begin when rated power is applied. After completion of each three hours of operation with rated power on, which shall include at least 360 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes.

4.4.3.3.2 Performance Characteristics to be measured - Upon completion of a maximum of 250 hours of cycling, the transmitter shall be checked for scale error. The tolerances for the reliability tests, Table I shall apply.

4.4.3.3.3 Failure criteria - Whenever performance characteristics fall below the acceptance requirements (4.4.3.3.2), at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused the other parts to fail.

4.4.3.3.4 Preventive maintenance - No preventive maintenance shall be attempted or accomplished on the samples while they are on test.

4.4.3.3.5 Disposition of samples upon completion of tests - Any sample tested for Reliability other than those tested under the Reliability Qualification Phase, may be delivered on contract provided it meets all the following requirements:

a. It is representative of production units currently being accepted.

b. It is in "good as new" condition or has been refurbished (see 6.3).

c. It is otherwise satisfactory.

4.5 Inspection conditions -

4.5.1 Standard conditions - Unless otherwise specified, all inspection required by this specification shall be made under the following conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

Temperature	Room ambient, 25° ±5°C.
Pressure	Normal atmospheric (approximately 29.92 inches Hg).
Humidity	Room ambient up to 90 percent relative humidity.
Input power	26 ±0.5V ac, 400 ±5 Hz.

4.5.2 Test readings - Unless otherwise specified before a test reading is taken, the transmitter shall either be tapped lightly or vibrated. If vibrated, the vibrator shall have a frequency within the 30 to 60 Hz. range and an amplitude within 0.0015 ± 0.0005 inch. Also, unless otherwise specified, two sets of readings shall be taken at each test point. One shall be taken by bringing the pressure up to, but not exceeding, each test point; the other shall be taken by bringing the pressure down to, but not below, each test point.

4.5.3 Attitude - Unless otherwise specified, the transmitter shall be tested in its normal operating position.

4.5.4 Test standards - The transmitter shall be tested against an indicating standard constructed of a standard production indicating unit selected for a maximum error of ±0.5 of a circular degree against a primary transmitting master. A primary transmitting master shall be a selected transmitting standard having a maximum inherent error of ±0.1 of a circular degree.

4.6 Inspection methods -

4.6.1 Examination of product - Each transmitter shall be examined to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests and other applicable specifications.

4.6.2 Scale error at room temperature - Prior to conducting the scale error at room temperature test, as an Individual test, the transmitter adjustment shall be turned to its limits in each direction and it shall comply with the requirements of 3.4. The transmitter adjustment shall then be adjusted, and locked, for operation with the test indicator and the adjustment shall not be further disturbed for the balance of the tests. The transmitter shall be tested and meet the scale error, at room temperature, at the test points specified in Table I for both upscale and downscale readings. The test points on the upscale reading shall be measured by bringing the input reading up to, but not exceeding, the specified test point. The downscale error shall be measured in the same manner, except that the input reading value shall be brought down to, but not below, the specified test point. When this test is conducted as an Individual test an indicator in accordance with MIL-I-7057 shall be connected as in Figure 2 and electrical zero and direction of rotation, i.e., upscale for increasing pressure at the transmitter, shall be verified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

4.6.3 Friction error - This test may be combined with the test specified in 4.6.2. The transmitter shall be tested for friction error at the test points specified in Table I. Two readings shall be taken; the first before and the second after the transmitter has been vibrated (4.5.2). The difference between the first and second readings shall not exceed the tolerances specified in Table I.

4.6.4 Position error - The position error tests may be conducted in conjunction with the test specified in 4.6.2. The transmitter, mounted in its normal operating position, shall be energized to obtain a reading of approximately 1/2 scale. Additional readings shall then be taken when the transmitter is in each of the following positions:

90° up, 90° down (longitudinal axis vertical).
90° CW, 90° CCW from vertical position above, (longitudinal axis horizontal).

The maximum change in indication from that in the normal operating position shall not exceed the tolerance specified in Table I.

4.6.5 Dielectric strength - A potential of 200V (root mean square) ac, at commercial frequency, shall be applied between isolated pins, and between pins and the case for 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the transmitter as a result of this test.

4.6.6 Sensors matching - (Differential pressure transmitters only)

(a) A pressure, relative to the atmosphere, equal to 1/5 of the range of the transmitter, shall be applied to the "P_2" port of the transmitter. A pressure shall be applied to the "P_1" port which shall exceed that applied to the "P_2" port by each of the test points listed in Table I, except any test point which requires that a pressure be applied to the "P_1" port the value of which, relative to the atmosphere, exceeds the value listed on the "Overpressure" row of Table I.

(b) The test of (a) above shall be repeated substituting "2/5" for "1/5" wherever it appears.

(c) The test of (a) above shall be repeated substituting "3/5" for "1/5" wherever it appears.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

TABLE I. Test points and tolerances for test conditions
(tolerances are plus or minus psi)

Range, psi	0-10	25	50	100	150	200	250	300	500	1,000	2,500	5,000
Test points, psi	0	0	0	0	0	0	0	0	0	0	0	0
	2	5	10	20	30	40	50	60	100	200	500	1,000
	4	10	20	40	60	80	100	120	200	400	1,000	2,000
	6	15	30	60	90	120	150	180	300	600	1,500	3,000
	8	20	40	80	120	160	200	240	400	800	2,000	4,000
	10	25	50	100	150	200	250	300	500	1,000	2,500	5,000
Overpressure, psi	20	50	100	200	300	400	500	600	750	1,500	3,000	6,000
Tolerances, psi												
Scale error at room temperature	0.2	0.5	1.0	2.0	3.0	4.0	5	6	10	20	50	100
Friction error 1/	0.1	0.2	0.5	1.0	1.5	2.0	3	3	5	10	25	50
Position error	0.1	0.3	0.5	1.0	1.5	2.0	3	3	5	10	25	50
Sensors matching (for differential transmitters only)												
1/5 Range	0.3	0.6	1.2	2.5	3.5	5.0	6	7	12	-	-	-
2/5 Range	0.4	0.7	1.5	3.0	4.5	6.0	7	8	15	-	-	-
3/5 Range	0.5	0.8	1.8	3.5	5.0	6.5	8	9	18	-	-	-
4/5 Range	0.6	0.9	2.0	4.0	5.5	7.0	9	10	20	-	-	-
Full range	0.7	1.0	2.3	4.5	6.0	8.5	10	11	23	-	-	-
Vacuum	0.3	0.6	1.2	2.5	3.5	5.0	6	7	12	-	-	-
High temperature	0.3	0.7	1.5	3.0	4.5	5.5	6	7	12	25	60	120
Low temperature	0.4	1.0	2.0	4.0	5.0	5.5	6	7	12	30	60	120
Vibration resonance	0.2	0.3	0.6	1.5	2.0	2.5	3	4	6	15	25	50
Vibration cycling	0.3	0.6	1.2	2.5	3.5	5.0	6	7	12	25	60	120
Reliability	0.4	1.0	2.0	4.0	6.0	8.0	10	12	20	40	100	200
Scale error overpressure	0.2	0.5	1.0	2.0	3.0	4.0	5	6	10	20	50	100

1/ This friction error applies wherever friction tests are specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

(d) The test of (a) above shall be repeated substituting "4/5" for "1/5" wherever it appears.

(e) The "P_1" and "P_2" ports shall be connected together and one pressure shall be applied to the two ports simultaneously. The value of this pressure, relative to the atmosphere, shall be slowly varied continuously from zero through the value listed on the "Overpressure" row of Table I, and back to zero. The indication shall not differ from zero at any time by more than the value called out in the Scale error "Sensors Matching - Full range" row of Table I.

(f) A vacuum of 14.0 $\pm$0.5 psi shall be applied to the "P_2" port of this transmitter. A pressure shall be applied to the "P_1" port which shall exceed that applied to the "P_2" port by each of the test points. The scale error shall not exceed the value listed in the Scale error "Sensors matching - Vacuum" row of Table I.

(g) For the 10-psi transmitters only, the test of (f) above shall be repeated substituting "7.5" for "14.0".

4.6.7 Proof pressure - For differential pressure transmitters in the range of 0-500 psi and all approximate gage pressure transmitters, a pressure of 75 psi, relative to the atmosphere, shall be applied to the case pressure release connection of the transmitter for 10 minutes. After removal of this pressure the transmitter shall be subjected to and meet the requirements of 4.6.8.

4.6.8 Sealing - Pressure and vent connections shall be closed with suitable plugs for this test. Each transmitter shall be immersed in a suitable deaerated liquid, such as water, within an enclosure. The absolute pressure of the air above the liquid shall be reduced to 2.5 inches Hg. Any bubbles coming from within the transmitter case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air in the various exterior parts of the case shall not be considered as a leak. A wetting agent shall be used in the liquid to facilitate removal of surface bubbles.

4.6.8.1 A leakage test by other methods, equivalent or superior in sensitivity to the test specified in 4.6.8, may be used upon approval by the procuring activity. Where a helium leak detector is employed for conducting this test, the leak rate shall not exceed 1 micron-cubic foot per hour at a pressure differential of approximately 1 atmosphere.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

4.6.9 Damping - (part numbers MS28131-7T, -8T, -11T, -12T, -23D, -23, -24D, -24, -25, -27D, and -27 only - The transmitter shall be connected to a suitable indicator and a pressure sufficient to produce a full-scale pointer deflection shall be applied to the transmitter. The test shall be conducted at 77°F using 1005 oil in accordance with MIL-L-6081 which has a measured viscosity of approximately 2,500 centistokes at -65°F. When the pressure is released suddenly, the interval of time required for the indicator pointer to travel from full scale to a point on the scale equivalent to 0.1 of the scale range shall be recorded. The time interval shall be within 2 to 4 seconds.

4.6.10 Suction - Gage pressure transmitters, with part numbers MS28131-2T, -4T, -6T, -20, -21, and -22, shall have a vacuum of 14.0 ±0.5 psi applied to the "P" port for 5 minutes with the "V" port open to atmosphere. The transmitter shall then be subjected to and shall pass the tests specified in 4.6.2 and 4.6.3.

4.6.11 Voltage and frequency variation - The transmitter shall be pressurized and a scale reading taken when the voltage and frequency are within the limits specified under 4.5. Under the same condition of pressure, the voltage and frequency shall be adjusted as listed and respective scale readings taken.

Percent volts	Percent frequency
90	95
110	105
90	105
110	95

No one of the latter four readings shall differ from the first reading by more than the tolerance specified in Table I for scale error at room temperature, for the pressure applied.

The test shall be conducted at three different pressure levels equally spaced over the transmitter range.

4.6.12 Seasoning - Each transmitter, in the ranges of 0-10 psi through 0-2,500 psi shall be subjected to 10,000 applications of pressure to produce an indication from zero to 80 ± 5 percent of the range of the transmitter, and back to zero at a rate of 30 ± 10 cpm. The 0-5,000 psi transmitter shall be subjected to 10,000 cycles of pressure from 2,500 psi to 3,500 psi at a rate of 40 ± 20 cpm. In each case, the pressure shall be essentially a displaced sinusoidal function of time. One ± 0.1 hour after the pressure cycling, the transmitter shall be tested for scale error at room temperature. The tolerances for scale error at room temperature and friction error, Table I, shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

4.6.13 High temperature - The transmitter shall be placed in a chamber and subjected to a temperature of 177 ±2°C for 4 hours. The transmitter shall be energized throughout the test. At the end of this period and with the transmitter still at this temperature, the transmitter shall be subjected to a scale error and friction error test. The scale and friction errors shall not exceed the tolerances specified in Table I for high temperature. Upon return to room temperature, the transmitter shall be subjected to and shall pass the tests specified in 4.6.2 and 4.6.3.

4.6.14 Low temperature - The transmitter shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502, at -54 ±2°C for a period of 4 hours after which it shall be subjected to the scale error and friction error tests while still at that temperature. The scale and friction errors shall not exceed the tolerances specified in Table I for low temperature. Upon return to room temperature, the transmitter shall be subjected to and shall pass the tests specified in 4.6.2 and 4.6.3. A fluid such as compass liquid in accordance with MIL-L-5020 shall be used during the low temperature test.

4.6.15 Vibration - The transmitter shall be subjected to vibration in accordance with MIL-STD-810 Procedure I, Method 514.1, equipment catagory (b), Curve E. The transmitter shall be electrically connected to an indicator with power and pressure applied.

4.6.15.1 Vibration resonance search and dwell - With the transmitter pressurized to mid scale, no condition of applied vibration shall cause vibration errors involving indicator pointer oscillation or variations in excess of the tolerance given in Table I for vibration resonance.

4.6.15.2 Vibration cycling - At the conclusion of the vibration cycling the transmitter shall be subjected to a scale error test and meet the tolerances of Table I, Vibration Cycling. No damage to the transmitter shall result from this test.

4.6.16 Low temperature exposure - The transmitter shall be subjected to a low temperature exposure test in accordance with MIL-STD-810, method 502. The transmitter shall be exposed to a temperature of -62° ±2°C for a minimum period of 48 hours after which it shall be operated throughout its entire range. It shall exhibit no erratic characteristics during operation at the -62° ±2°C temperature. The temperature shall then be raised to -54° ±2°C and the transmitter maintained at this temperature for a minimum of 24 hours. At the end of the 24-hour period, and while the transmitter is still at -54° ±2°C, the transmitter shall be subjected to the scale error and friction error tests. The scale error and friction error shall not exceed the tolerances specified in Tables I for low temperature. Upon return to room temperature, the transmitter shall be subjected to and shall pass the tests specified in 4.6.2 and 4.6.3. A fluid such as compass liquid in accordance with MIL-L-5020 shall be used during the low temperature exposure test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

4.6.17 High temperature exposure - The transmitter shall be placed in a chamber and subjected to a temperature of 177° ±2°C for 50 hours. The transmitter shall be energized throughout the test. At the end of this period and with the transmitter still at this temperature, the transmitter shall be subjected to a scale error and friction error test. The scale and friction errors shall not exceed the tolerances specified in Table I for high temperature. Upon return to room temperature, the transmitter shall be subjected to and shall pass the tests specified in 4.6.2 and 4.6.3.

4.6.18 Effect of lead length - The lead length between the transmitter and indicator synchro stators shall be changed from 5 to 300 feet of No. 20 AWG stranded copper wire. The indication shall not change more than 1° of arc. This test shall be conducted at room temperature.

4.6.19 Radio noise suppression - The transmitter shall be tested and shall meet the requirements of MIL-T-5350.

4.6.20 Electrical characteristics - The transmitter shall be tested and shall meet the requirements of MIL-T-5350.

4.6.21 Overpressure - A scale error test at room temperature shall be conducted just prior to this test and the readings recorded.

4.6.21.1 All transmitters - "P_1" and "P" ports - The pressure listed in the "Overpressure" row of Table I, relative to the atmosphere, shall be applied to the "P_1" or "P" port of the transmitter for 10 minutes, with the "P_2", "V", "V & CPR" port, if the transmitter has one, open to the atmosphere. One ±0.1 hour after the application of this pressure, the transmitter shall be tested for scale error on increasing pressure only. The readings shall not differ from the readings recorded immediately preceding this test by more than the value listed in the "Scale error overpressure" row of Table I. During this test, the presence of the stops as specified in 3.4.6.1 and 3.4.6.2 shall be verified and failure to comply with these paragraphs shall be cause for rejection.

4.6.21.2 Differential pressure transmitters - "P_2 port - The pressure listed in the "Overpressure" row of Table I, relative to the atmosphere, shall be applied to the "P_2" port of the transmitter for 10 minutes, with the "P_1" port open to the atmosphere. One ±0.1 hour after the application of this pressure, the transmitter shall be tested for scale error on increasing pressure only. The readings shall not differ from the readings recorded immediately preceding this test by more than the value listed in the "Scale error overpressure" row of Table I. During this test, the presence of the stops as specified in 3.4.6.1 and 3.4.6.2 shall be verified and failure to comply with these paragraphs shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

4.6.21.3 Gage pressure transmitters - "V", "V & CPR" port - A pressure of 75 psi, relative to the atmosphere, shall be applied to the "V", "V & CPR" port of the transmitters up to and including 0-50 psi for 10 minutes, with the "P" port open to the atmosphere. One ±0.1 hour after application of this pressure, the transmitter shall be tested for scale error on increasing pressure only. The readings shall not differ from the readings recorded immediately preceding this test by more than the value listed in the "Scale error overpressure" row of Table I. During this test, the presence of the stops as specified in 3.4.6.1 and 3.4.6.2 shall be verified and failure to comply with these paragraphs shall be cause for rejection.

4.6.22 Salt fog - The salt fog test shall be conducted in accordance with MIL-STD-810, method 509, at room temperature for a period of 48 hours. The transmitter shall be examined and there shall be no evidence of external corrosion or deterioration that would affect subsequent operation. The transmitter shall then meet the test specified in 4.6.5 and 4.6.8. Before the dielectric strength test is conducted, the external surface of the transmitter may be washed with distilled water and air dried. The zero adjustment mechanism shall then be tested for compliance with 3.4.

4.6.23 Case pressure release and test for helium - The pressure blow-out device shall be tested either under simulated increasing case pressure, or by actual injection of pressure into the transmitter case. The device shall operate at a pressure of 90 ±30 psi relative to the atmosphere. The transmitter shall be subjected to the test for helium by means of a mass spectrometer-type helium detector. Failure to detect helium shall be cause for rejection.

4.6.24 Case proof pressure - After sealing the openings made in 4.6.23, the inside of the transmitter case shall be pressurized at 150 psi (gage) for 5 minutes. Any leaks in, or distortion of, the case shall be cause for rejection.

4.6.25 Internal examination - The case of the transmitter shall be opened and the internal mechanism examined. Any deterioration or damage, which could in any manner prevent the transmitter from meeting operation or maintenance requirements during service life, shall be cause for rejection.

4.6.26 Torque - The transmitter shall be tested and shall meet the requirements of MIL-T-5350.

4.7 Rejection and retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and action taken to correct the defects shall be furnished the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The transmitter shall be packaged in accordance with MIL-STD-794, level A or C, as specified in contract or order (see 6.2). For level A packaging, the method of preservation shall be in accordance with MIL-P-116, Method IA, without preservative compound. When specified for the Army (see 6.2), if the unit container selected for Level A packaging is fiberboard, it shall conform to PPP-B-636, weather-resistant class.

5.2 Packing - The transmitter shall be packed in accordance with MIL-STD-794, level A, B, or C as specified in contract or order (see 6.2). When specified for the Army (see 6.2), the Level A container shall conform to PPP-B-601, overseas type, surface-treated in accordance with the requirements of the specification.

5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-794.

5.3.1 Precautionary marking - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container, wherever practicable:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The pressure transmitters are intended for use with pressure indicators conforming to MIL-I-6856, MIL-I-23479, MIL-I-7057 or other suitable remote, synchro pressure indicators for indicating aircraft pressures remotely, for high temperature installation.

6.2 Ordering data - Procurement documents should specify:

(a) Title, number, and date of this specification.

(b) The quantity, MS part number, and type number, if any, of the transmitter desired (see 1.2).

(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.

And for Army contracts

(1) When fiberboard unit containers shall conform to PPP-B-636 (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

(2) When shipping containers shall conform to PPP-B-601 (see 5.2).

(d) The laboratory that will conduct tests (see 4.4, 4.4.2.2).

(e) Reliability Test Phases (see 4.4.3).

6.3 Definitions -

6.3.1 Amplitude - Whenever the word "amplitude" is used, it shall mean the complete extent of motion as measured from one extreme to the opposite extreme.

6.3.2 Hermetic seal - A hermetic seal shall be a perfectly closed and air-tight seal, made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals which are dependent upon gaskets. The use of gaskets or O-rings for hermetic sealing is prohibited.

6.3.3 Back plate - The back plate is the surface of the case which is shown on MS28131 as having the pressure and vent connection bosses and the external zero reset.

6.3.4 Good as new - "Good as new" shall mean transmitters operated with less than 10 percent of the specified MTBF operation.

6.3.5 Refurbished - "Refurbished" shall mean that the transmitter has been completely overhauled with all component parts meeting current part standards; and the transmitter shall have been subjected to and met all the requirements of a new transmitter.

6.3.6 Corrosion resistance - The terms "corrosion resistant" and "protected against corrosion" as used in this specification shall mean a material which exhibits no visual evidence of corrosion subsequent to the Salt Fog Test, Method 509 of MIL-STD-810. The material shall be cleaned by ultrasonic methods or equivalent prior to testing to determine corrosion resistance.

6.4 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and directed to Provisions Governing Qualifications, DOD Document SD-6. Manufacturers are urged to arrange to have the products that they propose to offer to the Federal

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

Government tested for qualification in order to be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity.

6.5 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract: The contract shall have precedence over any specification.

(2) This specification: This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications: Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 Unless otherwise specified, for Air Force procurement only, the sampling plan B inspections (see 4.4) shall be performed at the suppliers facility or at a commercial laboratory acceptable to the Government, in lieu of at a Government laboratory. Test reports showing quantitative results for sampling plan B inspections and the samples tested therein shall be forwarded to the Government agency designated by the procuring activity for approval.

6.7 Marginal indicia - Asterisks are not used in this revision to identify the changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - AS
Air Force - 11
Army - AV

Reviewer activities:
Navy - AS
Air Force - 71
Army - AV

User activity
Navy - CG

Preparing activity:
Navy - AS
(Project No. 6620-0249)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-7748E TRANSMITTERS, PRESSURE, SYNCHRO, AIRCRAFT .320-DEGREE MOVEMENT, HIGH TEMPERATURE, (177°C), GENERAL PURPOSE

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:49Z
Check the source to verify that this is the current version before use.